Smart Research Assistant: Building Agentic AI for Automated Market Intelligence

Finsight AI

How do you invest today?

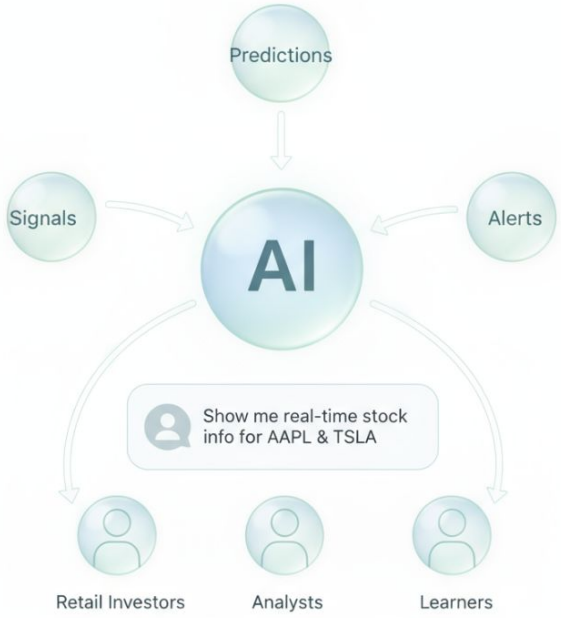
Use multiple apps for stock prices, charts, and news?

Spend hours tracking markets and analyzing data?

Struggle to get clear signals and explanations?

Do you know?

- AI can give **real-time predictions, signals, and alerts**.
- Natural-language queries make stock insights easier for everyone.
- Retail investors, analysts, and learners can all benefit from **AI-assisted decisions**.



The Challenge & Opportunity

Challenge	Opportunity
People have to use many apps and websites to track stocks and news	Unified **AI-powered platform** for all market needs
It takes a lot of time to collect and compare information.	Users can search any stock and instantly see live charts and key metrics.
Financial data is too technical, so non-experts find it confusing.	AI explains complex financial data in simple language that anyone can understand.
There’s no easy way to catch sudden market changes or trends in real time.	Real-time alerts, predictions, and comparisons help users make quicker, smarter choices.

Tech Stack

Frontend

React 18, TypeScript, Vite

Material-UI (MUI), Recharts, Axios

Native WebSocket API

Backend

FastAPI, Uvicorn (ASGI), Pydantic

asyncio

Google Generative AI (Gemini 2.5 -

Hugging Face

Transformers (dslim/bert-base-NER)

Prophet (time-series forecasting),

yfinance

Utilities / Build

SQLite (frentend tooling)

Realtime & Feeds

WebSockets (FastAPI WS),

aiohttp, feedparser

aiohttp

Data & Persistence

Logging (Python logging),

CORS middleware, datetime, hashlib

Node.js (frontend tooling),

Vite dev server, ESLint, Python 3.12

(venv)

Types / Contracts

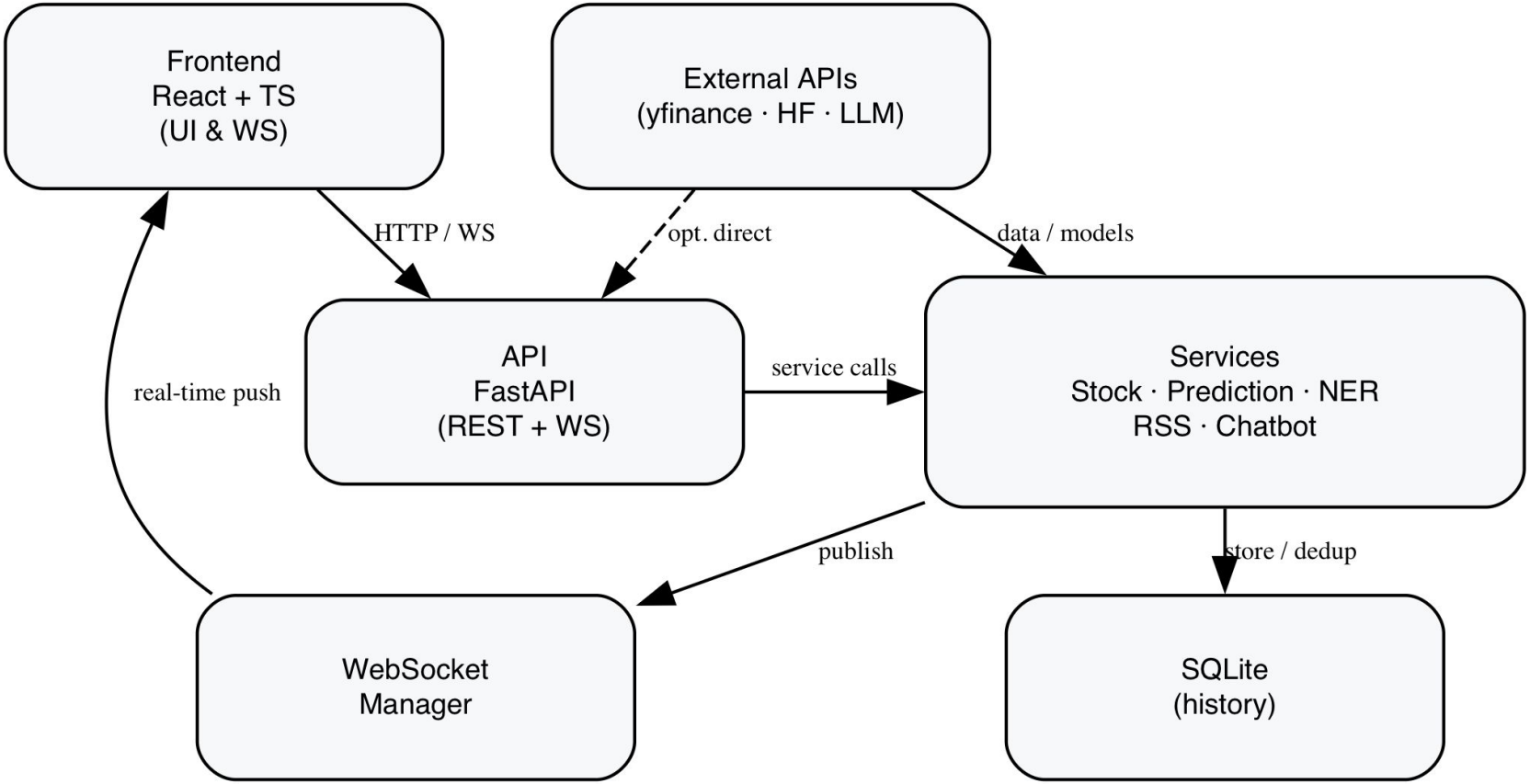
Pydantic schemas (webUI/backend/

models/schemas.py)

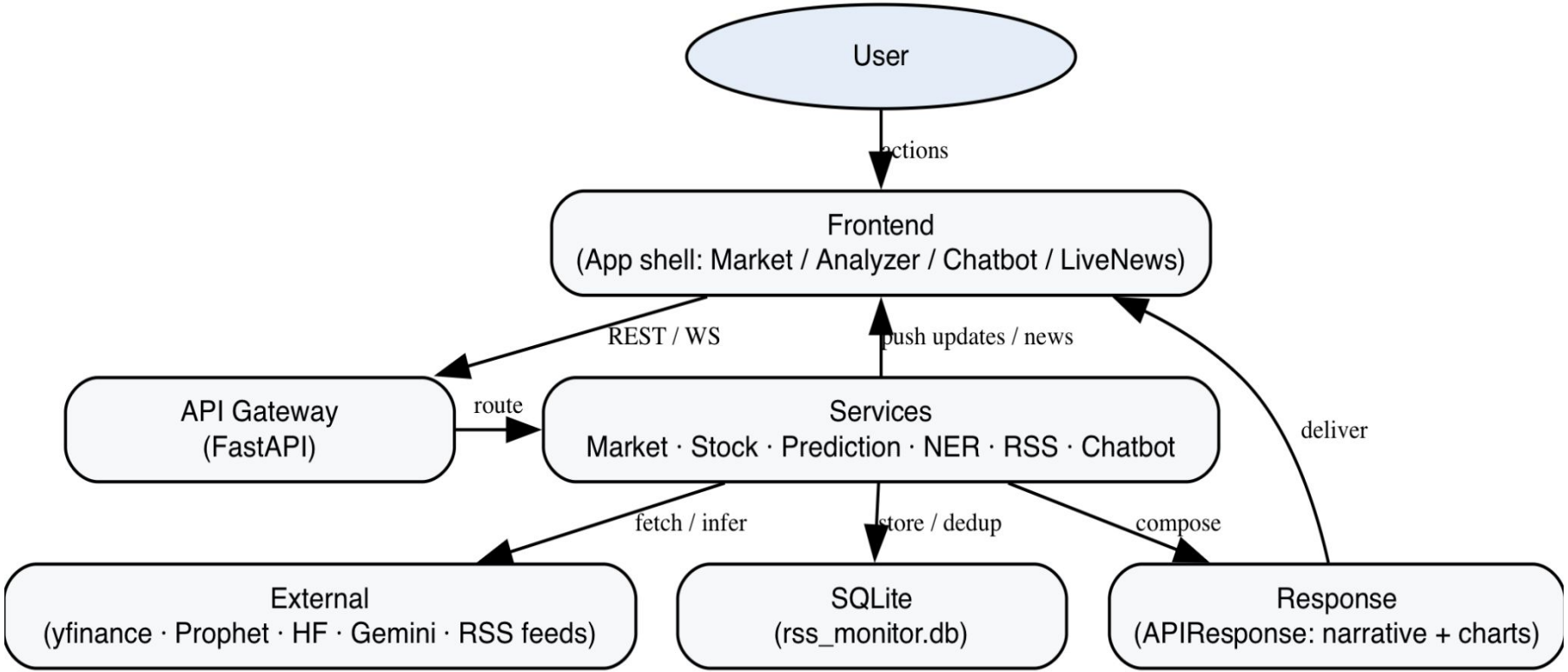
Unified API envelope: APIResponse

{ success, message, data

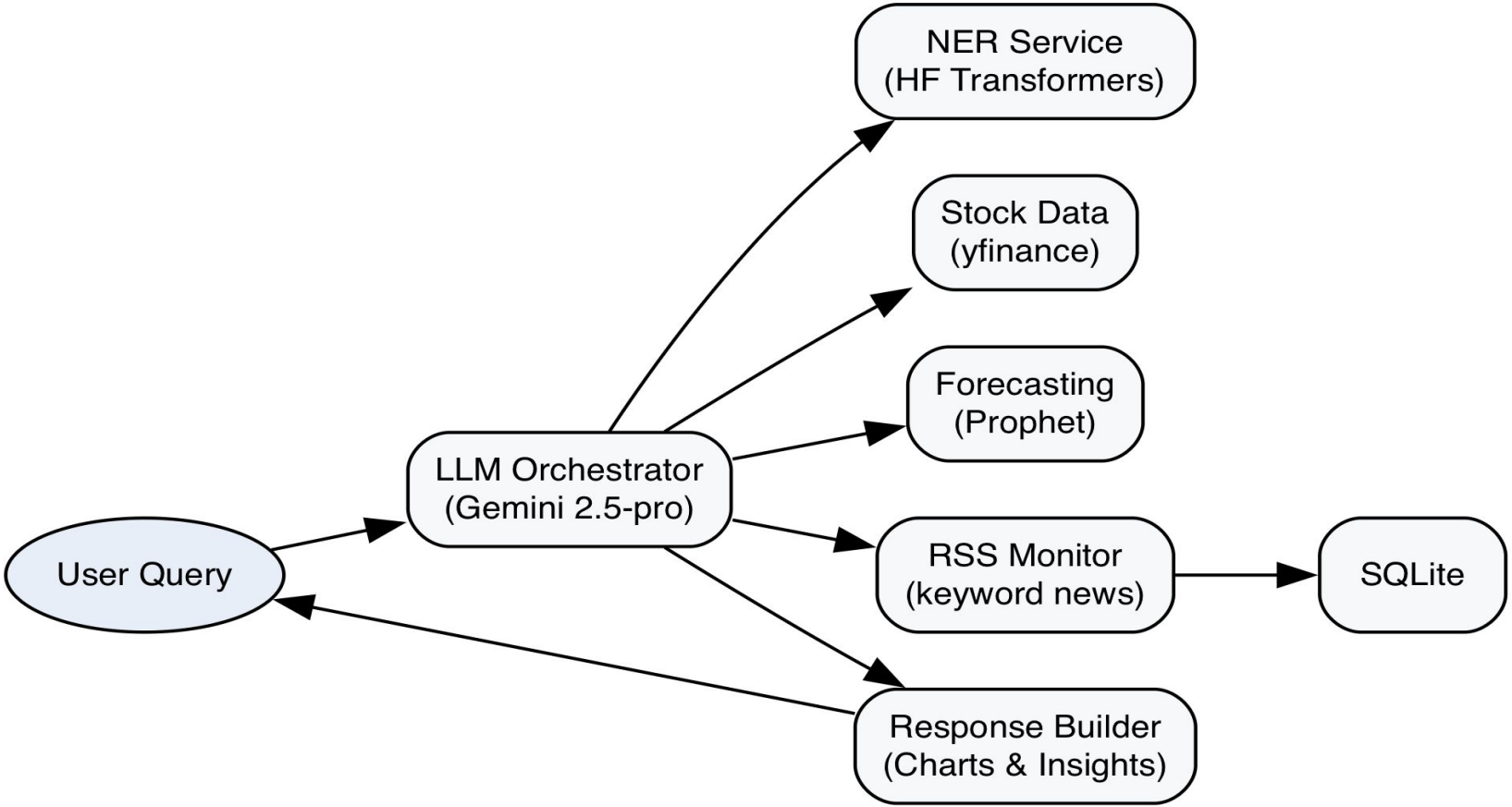
High Level Architecture



User flow


User
actions
Frontend
(App shell: Market / Analyzer / Chatbot / LiveNews)
REST / WS
push updates / news
API Gateway
(FastAPI)
route
Services
Market · Stock · Prediction · NER · RSS · Chatbot
deliver
fetch / infer
store / dedup
compose
External
(yfinance · Prophet · HF · Gemini · RSS feeds)
SQLite
(rss_monitor.db)
Response
(APIResponse: narrative + charts)

AI Architecture



Data Sources in Use

- **Yahoo Finance (yfinance)**
  - Real-time & historical stock prices (OHLCV)
  - Company info & fundamentals (market cap, P/E, EPS, dividend yield)
- **RSS Feeds (feedparser + aiohttp)**
  - Business & tech news (BBC, TechCrunch, The Verge, CNN, Reuters)
  - Headlines, summaries, publish dates, links
- **Open API**
  - AI chatbot responses & tool orchestration
  - Handles stock queries, news insights, forecasts

Prophet (Forecasting)

- Uses historical stock prices from Yahoo Finance
- Produces future trend and price predictions

SQLite Storage

- `rss_monitor.db` → prevents duplicate news entries
- Keeps minimal history for chatbot/news pipeline

FINANCIAL DATA ANALAZER

Problems Faced

- **Scattered Data:** Market info, news were in different places.
- **Lack of Context:** News didn’t explain why stocks or markets moved.
- **Hard to Use:** Non-experts couldn’t easily ask questions or get insights.
- **Slow Updates:** Charts and data were sometimes slow or outdated.
- **Prediction & UI Issues:** Forecasts needed fine-tuning; frequent chart updates slowed the app.

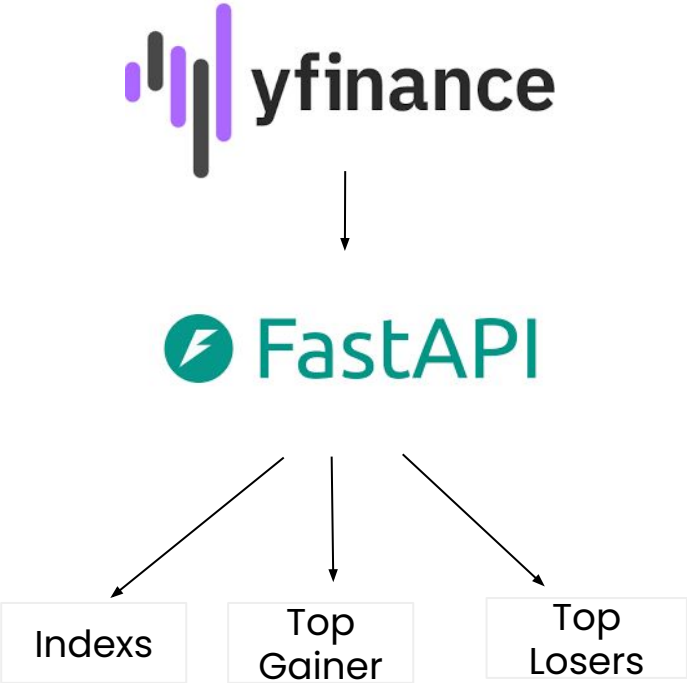
How We Solved It

- **All-in-One Platform:** Combined market data, news, charts, and predictions in one app.
- **Contextual News:** Linked news to stocks using AI, explaining *why* movements happened.
- **Chatbot:** Natural-language questions get answers with charts, predictions, and analysis.
- **Fast & Real-Time:** Optimized API calls and chart updates for smooth performance.
- **Better Predictions:** Tuned forecasting models with confidence intervals for accurate insights.

Market Summary

- **What it does:** View major indices and sectors with live news.
- **Updates:** Real-time every 30 seconds.
- **Value:** Quick market snapshot with drill-downs and context.

Market Summary

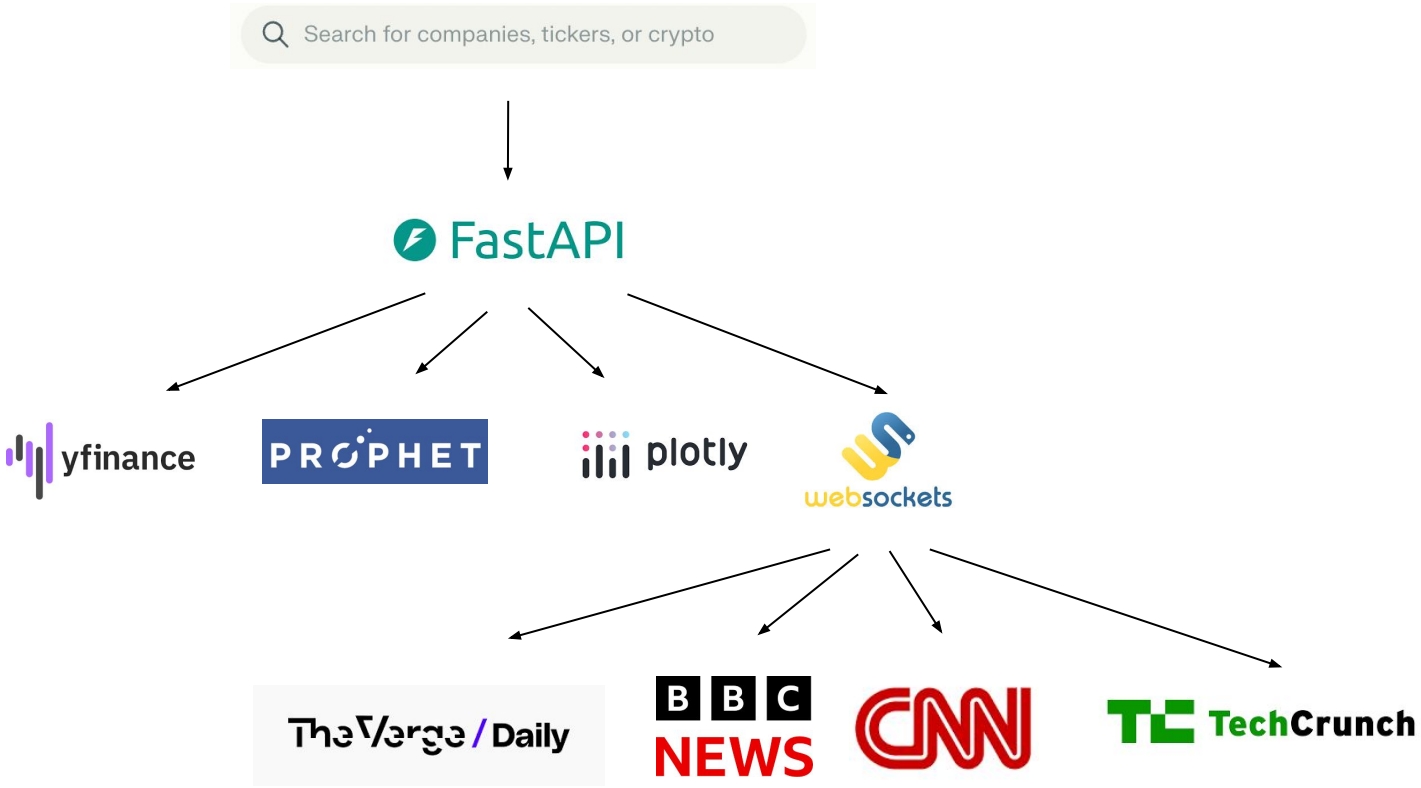


Stock Analysis

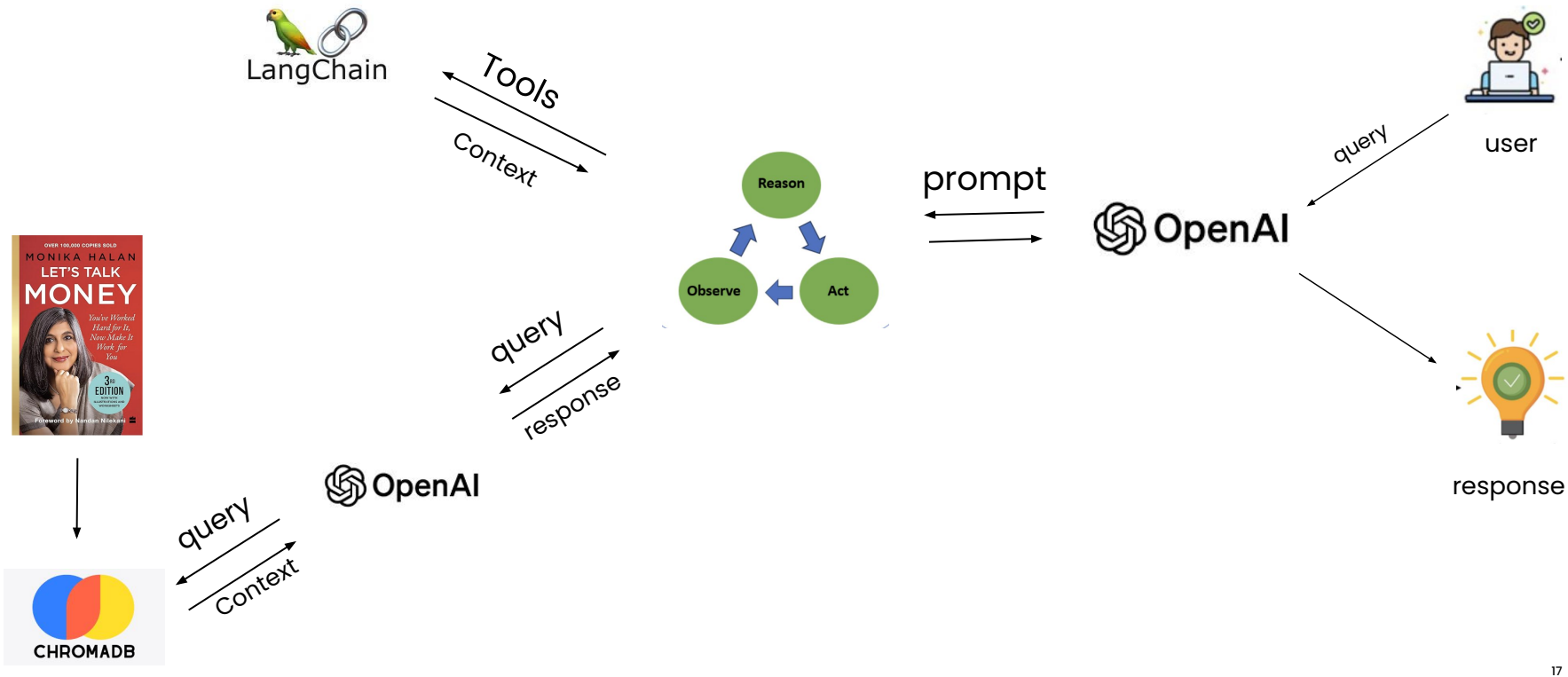
Stock Search & Time Series

- **What it does:** Search stocks by symbol/company with auto-suggestions.
- **Visuals:** Price, volume, technical indicators, and forecasts.
- **Value:** One-stop tool to explore, analyze, and forecast stocks.

Stock Analysis



Chatbot


LangChain
Tools
Context
Reason
Observe
Act
prompt
OpenAI
query
user
query
response
response
OVER 100,000 COPIES SOLD
MONIKA HALAN
LET'S TALK
MONEY
You've Worked Hard for It, Now Make It Work for You
3rd EDITION
Foreword by Nandan Nilekani
query
response
OpenAI
query
Context
CHROMADB

Tools

get_today_date

Return today's date

get_ticker

Lookup ticker symbol given company name

Finnhub api

get_news

Fetch news articles for a keyword

News api

get_fundamentals

Retrieve company fundamentals data

yfinance

get_reddit_posts

Fetch Reddit discussions by keywords

Reddit api

get_stock_price_history

Get historical stock price data

yfinance

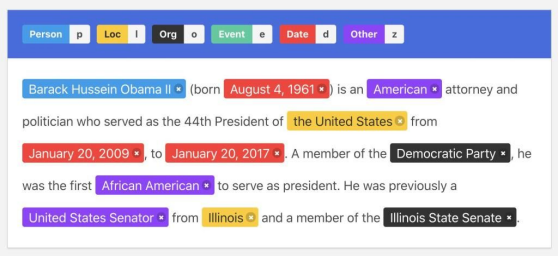
Personal_Finance_Assistant

QnA related to personal finance

RAG chatbot

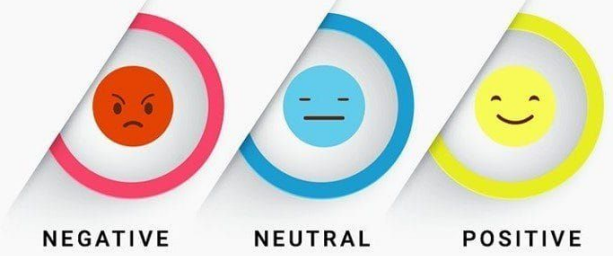
Models as Tools

NER



Google BERT

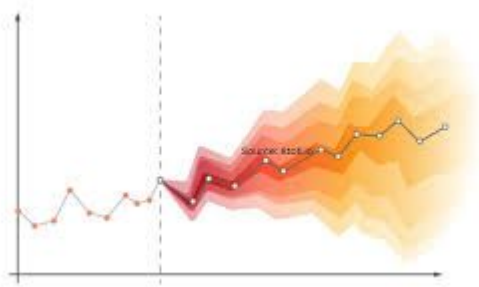
Sentiment



finBERT

(Financial sentiment Analysis)

Time Series



PROPHET

ReAct

- ReAct stands for "**Reasoning and Acting**."
- **Thought → Action → Observation**
- Helps AI provide context-aware and more accurate responses for complex tasks.

- Key Features:
  - Combines verbal reasoning with actions
  - Iterative process enhancing problem-solving, decision-making

🧠 Reasoning Process:

Step 1: 🤔 **Thought:** Thought: To predict the stock prices of Apple for next week, we need to generate stock price forecasts for the ticker symbol of Apple for the upcoming week.

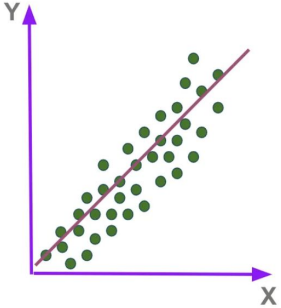
Action: get_ticker Action Input: "Apple" 🔧 **Tool:** get_ticker 📊 **Result:** AAPL

Step 2: 🤔 **Thought:** Action: stock_forecast Action Input: {"ticker": "AAPL", "periods": 7} 🔧 **Tool:** stock_forecast 📊 **Result:** 7-day forecast for AAPL: Date: 2025-09-23, Predicted: 241.33, Lower: 231.26, Upper: 251.25 Date: 2025-09-24, Predicted: 241.68, Lower: 230.91, Upper: 252.17 Date: 2025-09-25, Predicted: 242.13, Lower: 232.72, Upper: 252.32 Date: 2025-09-26, Predicted: 242.52, Lower: 233.38, Upper: 253.06 Date: 2025-09-27, Predicted: 243.35, Lower: 234.04, Upper: 253.47 Date: 2025-09-28, Predicted: 243.79, Lower: 233.91, Upper: 253.60 Date: 2025-09-29, Predicted: 244.04, Lower: 234.38, Upper: 253.99

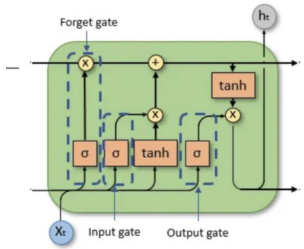
```python
custom_prompt = PromptTemplate(
    input_variables=["input",
                     "tools",
                     "tool_names",
                     "agent_scratchpad",
                     "chat_history"],

    template="""
        You are a smart market research assistant.
        Use the tools provided to answer user queries about
        stocks, companies, financial news, and market trends.
        Always keep the final answer comprehensive and detailed.
        Always mention news headlines with dates and sources
        and their sentiments as well.

        TOOLS:
        {tools}
        You may call any of these tools as needed.

        User's conversation history:
        {chat_history}
        When answering, always follow this format:

        Question: the user question
        Thought: your reasoning about what to do next
        Action: the name of the tool to call, one of [{tool_names}]
        Action Input: the input for the tool
        Observation: the result of the tool call
        ... (this Thought/Action/Observation can repeat as needed)

        Thought: I now know the final answer
        Final Answer: your complete and detailed answer to the user

        Begin!

        Question: {input}
        {agent_scratchpad}
        """
    )
```

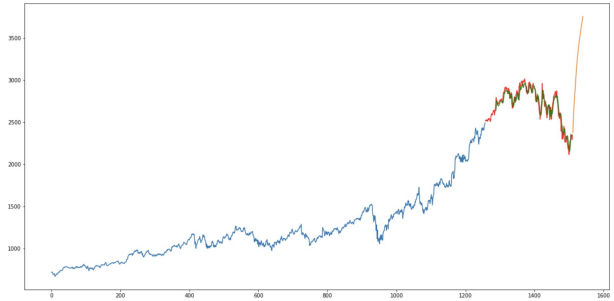
Time Series Modelling

Linear Regression

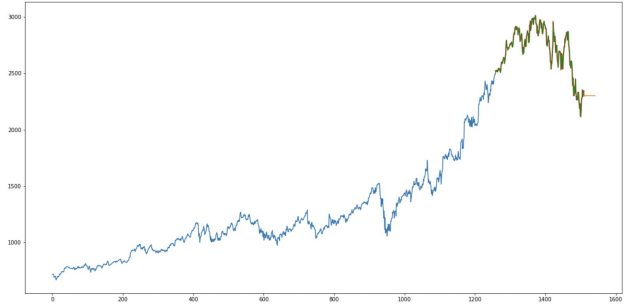


LSTM





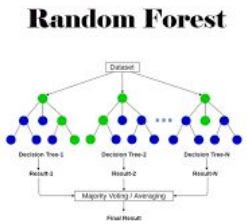
ARIMA

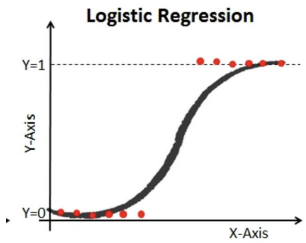


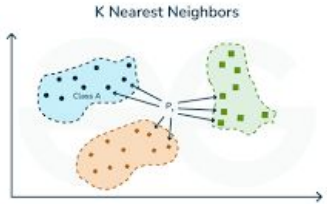
PROPHET

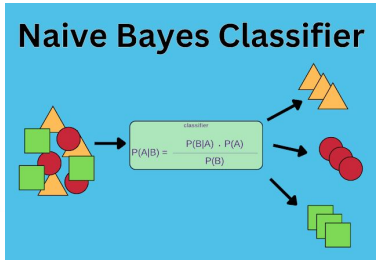
NER and Sentiment Analysis Modelling

kaggle +









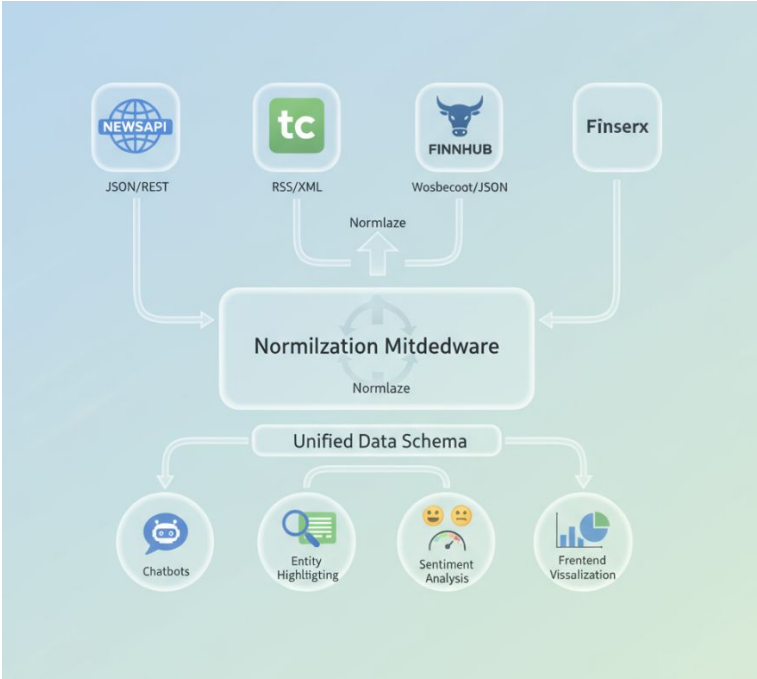
Google BERT

finBERT

(Financial sentiment Analysis)

1. API Keys and Ticker Symbols Mapping

2. Normalizing Multi-API News Data Formats



3. Managing GDELT Data Complexity & Multilingual Challenges

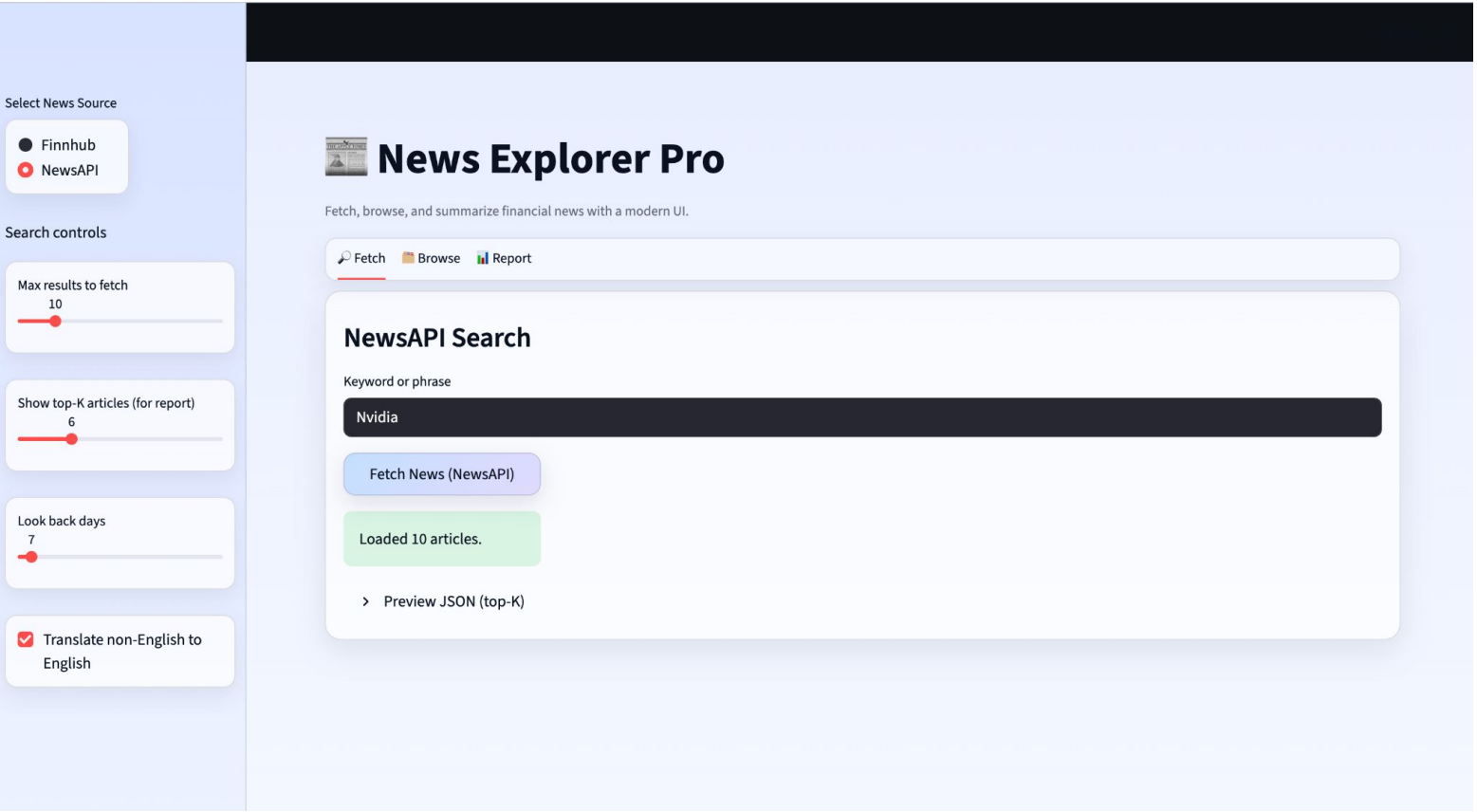
- **Challenge:** GDELT provided massive, multilingual news data with unpredictable structure.
- Translating and filtering non-English news for sentiment and entity analysis was difficult and error-prone.

4. Gemini Integration Problems

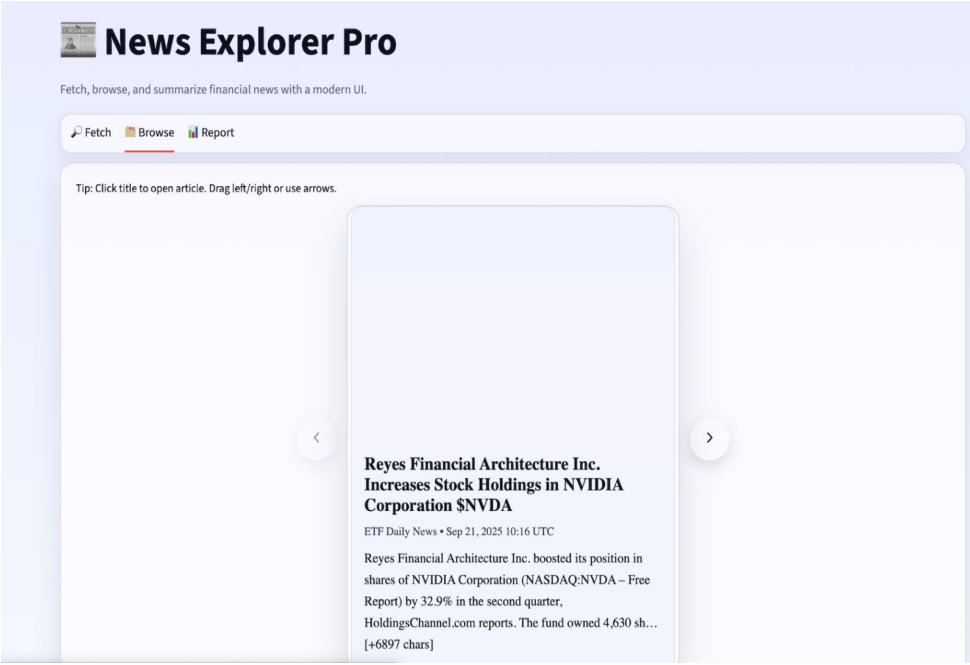
- **Challenge:** When using Gemini with LangChain, the chatbot called external tools but did not utilize their outputs, relying instead on Gemini's own predictions to generate responses. This caused a mismatch in expected tool usage and reduced chatbot effectiveness.

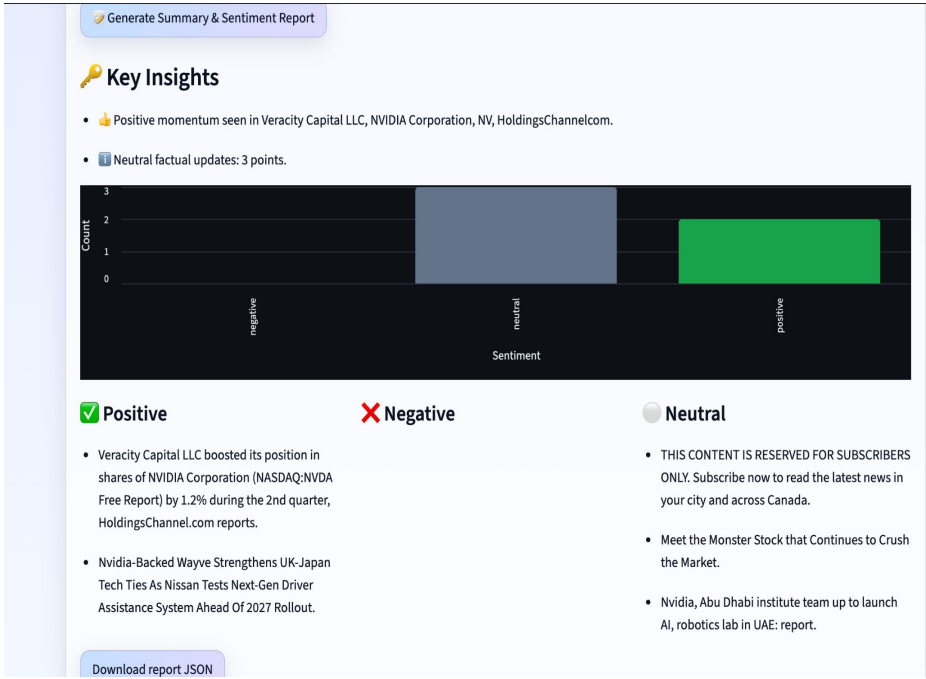
- **Solution:** Switched the underlying language model from Gemini to OpenAI within LangChain. OpenAI models properly invoked and integrated external tool outputs, restoring seamless tool functionality and improving chatbot response accuracy.

Interactive news dashboard



Interactive news dashboard





Future Features:

- **Events & Alerts:** Show earnings, dividends and send price/news alerts based on set configuration.
- **Advanced Multi-Language News Integration:** Expand multilingual support in news feeds with improved real-time translation.
- **Enhanced Chatbot Capabilities:** Integrate voice input and natural language voice responses.

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